

Question	Marking Guidance	Mark	Comments
02.1	Accept any two of the following for 1 mark; Volume/mass soda lime Concentration of soda lime Number/mass of woodlice Age of woodlice Same woodlice Species/type of woodlice Time for woodlice to acclimatise to (water bath) temperature (Starting) concentration of oxygen (inside the apparatus) Light intensity	1 (1 x AO3)	Accept weight/volume of woodlice for number/mass of woodlice Ignore 'amount' for concentration/volume/weight/mass

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02.2	1. Open the (3-way) tap; 2. Push/press the syringe (down);	2 (2 x AO3)	1. Accept descriptions of opening the tap, eg push/move/turn 2. Accept descriptions of the pushing the syringe down, eg apply pressure to the syringe

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02.3	1. No woodlice and all other conditions/apparatus/equipment the same; 2. To show that (respiring) woodlice are causing the drop to move OR To show that (respiring) woodlice are taking up the oxygen OR To show that (respiring) woodlice are causing the change in volume/pressure;	2 (2 x AO3)	1. Accept other inert objects instead of woodlice eg glass beads 1. Accept dead woodlice for no woodlice 1. Accept descriptions of 'all other conditions/apparatus/equipment the same' eg same investigation 2. Accept to show that (respiring) woodlice are affecting the results 2. Accept 'no other factor(s)' OR no other named factor OR 'nothing else' for woodlice

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02.4	Correct answer of 0.11 = 3 marks;;; 0.10816247795/0.1082600897/0.1082173395 = 2 marks (answer not to 2 dp) Evidence of $\frac{3.14 \times 1.25^2 \times 25}{(5 \times 60) \times 3.78}$ and incorrect answer to 2 dp = 2 marks (for input error into calculator) 0.41 = 2 marks (mean rate in $\text{mm}^3 \text{s}^{-1}$ for 3.78 g of woodlice) 32.45/32.48/32.47 = 2 marks (mean rate in $\text{mm}^3 \text{g}^{-1}$ in 5 minutes) 0.43 = 2 marks (used diameter instead of radius in calculation) Evidence of $\frac{3.14 \times 1.25^2 \times 25}{(5 \times 60) \times 3.78}$ and incorrect answer to not to 2 dp = 1 mark (for input error into calculator) 122.65625/122.767857142/122.718463 = 1 mark (volume of oxygen taken up in 5 minutes) 0.40885416/0.40922619047/0.4090615434 = 1 mark (mean rate in $\text{mm}^3 \text{s}^{-1}$ for 3.78 g of woodlice and not to 2 dp) 32.44874339/32.47826909/32.46520186 = 1 mark (mean rate in $\text{mm}^3 \text{g}^{-1}$ in 5 minutes and not to 2 dp) 0.4326499118/0.43286935813/0.43304358780 = 1 mark (used diameter instead of radius in calculation and not to 2 dp) 0.08 = 1 mark (distance travelled by the drop per second) 0.02 = 1 mark (just used 25 mm distance travelled by the drop and not worked out volume)	3 (3 x AO2)	For 3 marks accept 1.08×10^{-1}

